PRODUCT & ENGINEERING BLUEPRINT

Global Lead Intelligence & Sales Automation SaaS

A detailed feature catalogue, system architecture, connector strategy and phased delivery roadmap for building a global, multi-tenant lead intelligence and sales platform.

Planning baseline: project history shared in this conversation plus current official provider and security documentation.

Product constraints: free-first, connector-driven, tenant-safe, provenance-preserving and transparent about provider limitations.

Document date: 3 November 2026

STATUS DISCLAIMER

This is a forward-looking blueprint, not a claim that every feature is implemented. Historical implementation statuses are based on prior project reports and must be verified against the actual repository, tests and deployment before being treated as production-ready.

How to read this document

Label	Meaning
Reported implemented	Previously reported by development workflow; not independently verified in this PDF.
Under review / in progress	Prepared or underway; code review, integration or approval remains.
Planned	Intended product capability, not yet claimed to exist.
Research-dependent	Requires provider access, legal/policy review, pricing/quota check or technical proof.
Deferred	Intentionally postponed; must remain visible as a future release gate if security-critical.

Contents

- 1. Product definition and outcomes
- 2. Customers, roles and use cases
- 3. Product principles and boundaries
- 4. End-to-end user journey
- 5. Architecture and canonical data model
- 6. Discovery and lead intelligence feature catalogue
- 7. CRM, sales automation and outreach feature catalogue
- 8. Connector framework and global source strategy
- 9. Security, privacy and governance
- 10. SaaS operations, subscriptions and administration
- 11. Eighteen-phase roadmap
- 12. Quality, testing and release gates
- 13. Free-first cost strategy
- 14. Risks, decisions and launch criteria
- 15. Research references and project baseline

1. Product definition and outcomes

Working definition: a multi-tenant SaaS platform that helps businesses discover potential customers, combine business information from permitted sources, assess and organize prospects, manage outreach, automate follow-ups and measure conversion outcomes in one workspace.

The product is not simply a Google Maps scraper or contact exporter. Its differentiation is a unified global lead-intelligence and sales workflow: multiple source connectors feed one normalized lead record, and that same record moves through qualification, CRM, outreach, follow-up and conversion tracking.

Core value proposition

- **Discover:** search by country, geography, industry, keywords, company attributes and supported source.
- **Unify:** merge duplicate observations into one canonical lead while retaining source evidence.
- **Understand:** enrich with permitted business facts, provenance, freshness and confidence.
- **Prioritize:** filter, segment and score prospects using explainable criteria.
- **Execute:** assign owners, manage pipeline stages, record interactions and schedule follow-ups.
- **Convert:** run approved campaigns and sales tasks, then record outcomes.
- **Measure:** report on discovery, data quality, activity, pipeline movement and conversion.

What it is not

- Not permission to scrape every website or social platform. Every source needs an approved access method and permitted use.
- Not a guarantee that every record has phone, email, website or a named decision-maker.
- Not an autonomous spam engine. Outreach must respect consent, local marketing law, suppression lists and platform rules.
- Not a promise that every API is free at commercial scale. Free access, free development and free redistribution are different.

2. Customers, roles and use cases

Customer	Job to be done	Useful features
Local service businesses	Find relevant nearby businesses and manage prospecting.	Country/city/category discovery, lead lists, reminders, simple pipeline.
Agencies / freelancers	Build qualified prospect lists for services.	Multi-source discovery, dedupe, exports, qualification, outreach tracking.
B2B sales teams	Research accounts and manage sales activity.	Firmographic enrichment, segments, pipeline, collaboration and reports.
Recruiting / staffing	Identify organizations in target sectors.	Company discovery, filters and account notes; people data only where permitted.
Exporters / suppliers	Find businesses in international markets.	Country registries, local identifiers, region filters and provenance.
Small organizations	Run sales without a dedicated CRM/IT team.	Guided setup, templates, simple workflow and low-cost usage.

Roles

- **Platform operator:** platform-wide provider configuration, abuse controls and system health; separate from customer permissions.
- **Tenant owner:** organization settings, members, credentials, subscription and data export/deletion.
- **Tenant admin:** member and connector management within assigned permissions.
- **Sales manager:** lead assignment, pipeline configuration, activity and reports.
- **Sales representative:** prospecting, qualification, assigned leads and follow-up.
- **Analyst/read-only:** view authorized leads and reports without editing.

Boundary: a tenant is the customer workspace; a user is a person belonging to it. Leads, notes, activities, tasks, credentials, files, campaigns and reports must be tenant-scoped. Platform-owned provider credentials must be explicitly distinguished from customer-owned credentials.

3. Product principles and boundaries

- **Free-first:** prefer open data, open-source tools and customer imports; no paid API/service/dependency without an explicit cost and approval review.
- **Connector-first:** provider-specific logic stays behind a stable adapter contract.
- **Tenant isolation:** derive tenant identity from authenticated server context, never trust a client-supplied tenant ID.
- **Truthful data:** never fabricate contact details, integration status, verification or result counts.
- **Provenance:** preserve source, source record ID, retrieval time and license/attribution metadata.
- **Safe dedupe:** preserve evidence and history; fuzzy matches should be reviewed rather than silently merged.
- **Human control:** users approve outbound content, sender, audience and automation rules.
- **Provider compliance:** use official APIs or expressly permitted datasets; do not bypass access controls or rate limits.
- **Observable failure:** distinguish not configured, unauthorized, rate-limited, unavailable, unsupported, partial and successful-empty.
- **Small-business usability:** usable without dedicated IT, data or compliance staff.

4. End-to-end product journey

1. **Workspace setup:** register, create tenant, invite team, choose target markets and categories.
2. **Connect sources:** enable supported connectors, configure credentials and review permissions/status.
3. **Discover:** define geography, industry, keywords, source and filters; run a provider-compliant search.
4. **Normalize:** map raw records into canonical lead structure; normalize identifiers and locations.
5. **Deduplicate:** match strong identifiers and create review candidates for fuzzy matches.
6. **Enrich:** add permitted business facts with source and freshness metadata.
7. **Qualify:** filter, tag, score and route leads for review.
8. **Manage pipeline:** assign, stage, annotate and link activities/tasks.
9. **Outreach:** use approved templates and compliant channels; record provider-confirmed sends.
10. **Follow up and learn:** create reminders, record outcomes and compare source/campaign conversion.

A provider may return only a name and registration number. That is a valid partial lead, not a failed record and never a reason to invent missing fields.

5. Architecture and canonical data model

Layer	Responsibilities	Design direction
Web application	Workspace UI, search, lead tables, CRM, reports, settings.	Authenticated APIs; no provider secrets in browser.
API/application	Auth, authorization, validation, tenant context, orchestration.	Thin routes; shared services and error mapping.
Connector framework	Adapters, credentials, quotas, retries, normalization hooks.	One contract; provider-specific logic isolated.
Lead intelligence	Canonical records, identifiers, provenance, dedupe, merge/undo.	Source-neutral model; retain evidence only as terms permit.
CRM/workflows	Pipeline, tasks, activities, assignments, campaigns.	Tenant-scoped state transitions and audit.
Persistence	PostgreSQL, migrations, indexes, backups.	Tenant-scoped records and explicit constraints.
Async/operations	Jobs, schedules, retries, quotas and observability.	Tenant-aware queues when needed.
Providers	Registries, maps, search, approved social APIs, email.	Treat responses as untrusted; honor terms.

Core entities

Entity	Purpose
tenants / users	Customer workspace and member identity, roles and status.
leads	Canonical business prospect and normalized business attributes.
lead_sources	Source observations, provider IDs, URL, retrieval time, license and attribution.
lead_identifiers	Typed normalized identifiers such as domain, phone, ABN or company number.
lead_merge_history	Survivor, merged record, actor, timestamp and undo metadata.
connector_credentials	Encrypted tenant-owned secrets; never plaintext in ordinary API responses.
connector_runs	Tenant, connector, filters, status, counts, timing and error category.
lists / segments	Saved collections and dynamic filters.
pipelines / stages	Tenant-configurable sales process.
activities / tasks	Calls, meetings, emails, notes, follow-up work and reminders.
campaigns / messages	Approved campaign configuration, templates and provider send IDs.
audit_events	Sensitive actions and business changes; no secrets.
subscriptions / usage	Plan, entitlements, counters and payment references if billing is enabled.

Canonical lead design rules

- Stable internal lead ID independent of provider IDs.

- One lead can have multiple source records and provider identifiers.
- Keep field provenance: source, timestamp, confidence/verification and user-edit state.
- Normalize international phones with country context; preserve display and normalized forms.
- Distinguish company website domains from social URLs and URL shorteners.
- Use typed country-specific registration identifiers.
- Automatically merge only on strong evidence; fuzzy matches go to human review.
- Never overwrite a verified/user-edited field silently.

6. Discovery and lead intelligence feature catalogue

6.1 Global discovery

Feature	Detailed behavior	Priority
Country selector	Supported country list; ISO codes internally; show source coverage.	MVP
Region/city filters	State, province, region, city or locality when provider supports it.	MVP
Industry/category	Map user categories to provider taxonomies; keyword fallback.	MVP
Keyword search	Company name, product/service terms or registry name.	MVP
Radius filters	Coordinates/radius only with compliant source and clear precision.	Later
Source selection	One or multiple compatible connectors; show limitations.	MVP
Search preview	Explain estimated coverage/limits without invented counts.	Later
Saved searches	Save filters and rerun manually or on permitted schedule.	Later
Discovery jobs	Queued/running/completed/partial/failed with counts and errors.	MVP
CSV/XLSX import	Column mapping, preview, validation, duplicate checks and row errors.	Reported; verify
CSV/XLSX export	Export selected permitted fields with authorization and audit.	MVP

6.2 Unified lead intelligence

Feature	Detailed behavior	Priority
Canonical normalization	Map every source into one schema.	Foundation
Source provenance	Show where facts came from and when retrieved.	Foundation
Identifier normalization	ABN, company number, domain, phone, email, provider ID.	Foundation
Duplicate detection	Exact strong identifiers; fuzzy candidate review.	Foundation
Merge and undo	Preserve sources/identifiers and record merge history.	Reported; verify
Field conflict handling	Prefer verified/current evidence under explicit rules.	MVP
Freshness indicators	Last checked, stale/unknown; refresh only when permitted.	MVP
Data quality score	Explain missing/conflicting/stale fields; no false certainty.	Later
Enrichment waterfall	Query eligible sources in order, respecting quotas and terms.	Later
Lead confidence	Explain signals from source reliability and identifier matches.	Later

6.3 Search and filtering

- Filter by country, region, city, category, keyword, source, status, owner, tags and dates.
- Filter by presence/absence of website, public phone/email, registration ID or social URL.
- Multi-select and saved views for recurring workflows.
- Sort by name, location, source, freshness, score, created date and stage.
- Use server-side pagination/filtering; do not load the full tenant database in the browser.
- Distinguish zero matches from connector failure.

6.4 Enrichment safeguards

Enrichment is opt-in. Each adapter declares lawful fields, required credentials, quota, geography, retention and attribution. Personal data about employees, owners or directors needs additional privacy/legal review. A public registry record is not automatic permission to use a person's details for marketing.

7. CRM, sales execution and automation

7.1 Lead workspace

- Lead detail with identity, location, registry IDs, website/domain, public contact channels if available, category, tags and owner.
- Source timeline with observations, imports, updates, merge history and enrichment attempts.
- Editable notes/custom fields with tenant ownership and audit history.
- Lead statuses: new, reviewed, qualified, disqualified, do-not-contact, converted or tenant-defined.
- Owner/team assignment and permission-checked bulk assignment.
- Secure attachments only after tenant-scoped file storage exists.
- People/contact records separate from company records; require lawful source and consent context.

7.2 Pipeline and opportunity

Feature	Behavior
Pipeline configuration	Owner/admin creates pipelines and ordered stages.
Stage movement	Drag/drop or actions; actor, old/new stage and timestamp recorded.
Opportunity value	Optional amount, currency, close date and probability.
Lead conversion	Convert qualified lead to opportunity without losing source/activity history.
Lost reasons	Configurable budget, timing, no response, not fit, competitor etc.
Forecast	Aggregate by stage, owner and close date; clearly label as estimate.
Team views	Assignments, overdue tasks, stage aging and activity volume.

7.3 Activities and follow-up

- Log calls, meetings, emails, messages, demos, proposals and manual touchpoints.
- Store time, owner, direction, outcome, notes and linked lead/opportunity.
- Manual logging works before email/telephony integrations exist.
- Keep provider message IDs and delivery states when connected.
- Never claim a message was sent unless provider confirms it.
- Tasks include due date, priority, owner, related lead, completion, snooze, reschedule and reassignment.
- In-app reminders first; external notifications only after a compliant provider is configured.

7.4 Automation engine

Capability	Required behavior / safeguards
Triggers	Lead created, stage changed, task overdue, inactivity, import completed, custom event.
Conditions	Country, category, source, score, stage, owner, tags and fields.
Actions	Create task, assign, tag, update allowed field, notify, enqueue approved outreach.
Delay	Scheduled action; tenant timezone and cancellation supported.
Approval	Human approval by default for outbound messages and bulk actions.
Suppression	Check do-not-contact, opt-out, invalid address and provider restrictions.
Execution history	Rule version, trigger, action, retries, actor and error; no secrets.
Limits	Tenant daily caps, provider quotas, concurrency and global safety controls.
Kill switch	Tenant and platform operator can pause automation.

7.5 AI-assisted sales agent — later

- Summarize a lead using permitted cited facts.
- Suggest qualification questions and next-best actions with reasons.
- Draft personalized outreach only from verified facts and approved templates.
- Summarize activity history and prepare meeting briefs.
- Suggest follow-up timing from explicit rules and observed outcomes.
- Human review before sending; label generated content and uncertainty.
- Never invent relationships, facts, claims or identity.
- AI provider, processing, retention, cost and tenant opt-in controls remain separate.

7.6 Campaign/outreach management — later

- Campaign setup, audience segments and exclusions.
- Templates, personalization tokens, preview and approval.
- Sender identity, scheduling and provider integration.
- Delivery, bounce and response tracking where provider exposes it.
- Unsubscribe, suppression and complaint handling.
- Campaign analytics and conversion attribution.
- Market-specific legal review before automated sending.

8. Connector framework and global source strategy

8.1 Standard connector contract

- Metadata: connector ID, name, category, supported countries, operations and docs URL.
- Configuration: required credentials, scopes, tenant-owned/platform-owned classification.
- Operations: search, verify, enrich and optional incremental sync; unsupported actions explicit.
- Normalization: provider payload to canonical lead plus source identifiers.
- Reliability: timeout, bounded retry, backoff, Retry-After, response-size limits and safe error mapping.
- Compliance: attribution, license, retention, commercial use, rate limits and prohibited use.
- Observability: status, last successful run, latency, counts, rate-limit state and actionable errors.

8.2 Connector inventory

Source	Role	Status / caveat
Google Maps / Places	Local business discovery and place identity.	Legacy endpoint reported preserved. Official Places policy restricts caching/storage and requires attribution; do not assume scraped content is compliant.
OpenStreetMap / Nominatim	Geographic/business discovery.	Reported connector; public service disabled by default in hardening. Public endpoint is not a bulk POI backend.
CSV / Excel	Customer-owned data import.	Reported implemented; verify mapping, validation, dedupe and export.
Brave / Bing / Google CSE	Web-search discovery.	Reported adapters; each has separate key, quota and terms.
Companies House	UK company registry.	Reported implemented; authentication required; published 600 requests per five minutes.
Brreg	Norway organisation registry.	Reported implemented; API and NLOD license documented.
ABN Lookup	Australian business number/name search and verification.	Prepared patch/review package; credential-store integration pending.
Latvia open data	Next possible registry.	Planned research candidate; verify API, license, quota, fields and commercial use.
Lithuania JARS / France SIRENE / Estonia RIK	Country registry expansion.	Research candidates; access, schema, rate and license need confirmation.
Japan NTA / Denmark CVR / Finland YTJ / Brazil CNPJ	Additional country coverage.	Research candidates; no implementation claim.
Instagram / Facebook	Authorized professional account/Page capabilities.	Research-dependent; permissions and asset ownership limit use.
LinkedIn	Approved self-service or partner products.	Research-dependent; many API permissions require approval; not a general prospect scraper.
YouTube	Channel/video discovery and analytics.	Research-dependent; API quota and authorized use cases apply.

8.3 Country connector delivery workflow

1. Select a country using customer demand, official data availability, legal clarity, quality and cost.
2. Find the official registry/data portal, not an unverified third-party list.
3. Document access model: open download, API key, registration, OAuth, paid tier or approval.
4. Record fields available/absent; never promise phone/email if not supplied.
5. Review license, attribution, commercial use, caching, retention, redistribution and privacy.
6. Build a small read-only proof of concept and bounded test query.
7. Normalize identifiers and preserve original provider record ID.
8. Test empty, partial, malformed, rate-limited and unauthorized responses.
9. Expose honest status and limitations; feature-flag if needed.
10. Measure match rate, duplicate rate, useful fields and reliability before expansion.

9. Security, privacy, governance and trust

9.1 Authentication and authorization

- Secure registration/login, password hashing, token expiry/revocation and account recovery.
- Tenant membership and roles: owner, admin, manager, representative, analyst and platform operator.
- Every object-level action checks resource and tenant ownership.
- Least-privilege provider scopes; MFA for privileged roles when feasible.
- Audit credential changes, exports, bulk actions, role changes, merges and deletion.

9.2 Tenant isolation

- Tenant identity comes from verified identity/membership, not request body or header alone.

- Every tenant-owned query, update, delete, cache key, background job and file object preserves tenant scope.
- Composite constraints and PostgreSQL Row-Level Security may provide defense in depth after connection-pooling tests.
- Negative cross-tenant tests for leads, exports, merges, credentials, tasks, files and async jobs.
- Tenant-aware quotas for shared provider and compute resources.
- Keep platform credentials separate from customer-owned credentials.

9.3 Secrets and credentials

- Dedicated credential store, authenticated encryption at rest, key versioning and out-of-band key backup/recovery.
- Never return plaintext secrets from status APIs or log authorization headers.
- Only owner/admin can add, replace, validate or delete credentials.
- Separate app secrets, tenant connector secrets and platform provider secrets.
- No silent process-global fallback in multi-tenant production.
- Provider validation stores status/time/error category, not the secret.

9.4 SSRF and unsafe URL handling

- User-supplied website URLs are an SSRF boundary.
- Validate DNS/IP targets; block loopback, private, link-local and metadata addresses; re-check redirects.
- Bound redirects, bytes, decompression, connection and total time.
- Allow only expected schemes and reject embedded credentials.
- Test DNS rebinding, IPv6, redirect chains and mixed DNS answers.

9.5 Privacy and marketing

- Document data categories, purposes, sources, legal basis and retention.
- Provide privacy notice, correction, export and deletion workflows as applicable.
- Treat named individuals and direct contact details as personal data where law requires.
- Maintain do-not-contact and opt-out suppression.
- Respect jurisdiction-specific email/electronic marketing rules and sender identification.
- A public registry listing is not consent to marketing.

9.6 Security framework

Use OWASP API Security Top 10 as a threat checklist: broken object/function authorization, broken authentication, property-level authorization, unrestricted resource consumption, SSRF, misconfiguration and unsafe third-party API consumption. Security is a release gate, not a final polish phase.

10. SaaS operations, subscriptions and administration

10.1 Onboarding and offboarding

- Create tenant and first owner atomically; provision default pipeline/settings.
- Guided setup: company profile, markets, categories, connector choices and import.
- Show whether connectors are open/free, require user credentials, approval or possible charges.
- Demo data/actions are explicitly labelled and isolated from real customer data.
- Offboarding suspends new operations, revokes sessions/credentials and follows retention/deletion policy.

10.2 Usage meters and entitlements

Meter	Possible unit	Controls
Discovery	Search jobs, provider calls, results processed.	Tenant quota, provider quota, daily cap.
Stored leads	Active canonical leads or records.	Plan cap, archive/export/delete.
Enrichment	Enrichment runs or provider calls.	Per-source budget and estimate.
Seats	Active team members.	Role and invitation controls.
Automation	Active rules, executions, queued actions.	Concurrency and execution budget.
Outreach	Messages sent via connected providers.	Consent, provider policy, daily cap, suppression.
Exports	Rows or export jobs.	Permission, audit and expiring download.

10.3 Billing — later

- Keep plan/entitlement logic separate from payment-provider code.
- Hosted checkout only after fees, country support, taxes and terms are reviewed.
- Handle trial, active, past due, canceled, grace and suspended states.
- Verify webhook signatures and process idempotently.
- Define fair data-export access on cancellation/downgrade.
- Do not promise free forever until real hosting/provider/support costs are measured.

10.4 Admin and observability

- Connector health: configured, not configured, healthy, degraded, rate-limited, unauthorized or disabled.
- Job dashboard: duration, found/imported/merged/rejected counts, retries and error category.
- System health: API latency, DB availability, queue depth, CPU/memory, error rate and backups.
- Tenant usage and abuse controls; restrict platform access to customer lead contents.
- Structured logs with redaction, correlation IDs and no unnecessary personal data.
- Incident runbooks for credential leak, tenant exposure, provider ban, key loss and restore.

11. Eighteen-phase implementation roadmap

The 18-phase roadmap preserves the previously discussed direction. Phases 1–7 and selected connector work are reported as implemented; Phase 8 security remediation is partially deferred; Phase 9 remains active. These are status reports, not fresh repository verification.

Phase 1 — Multi-tenant foundation and security hardening

Purpose: Make a single-user tool safe as a shared SaaS.

Status: Reported implemented; verify.

Feature scope:

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Dependencies: Database and server foundation.

Exit criteria: Two-tenant tests prove isolation; production configuration fails closed.

Phase 2 — Unified lead intelligence

Purpose: Replace fragmented business rows with canonical leads.

Status: Reported implemented; verify.

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Dependencies: Phase 1.

Exit criteria: Migration recovery documented; dedupe and cross-tenant tests pass.

Phase 3 — Connector framework

Purpose: Add sources without rewriting core product.

Status: Reported implemented; verify.

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Dependencies: Canonical lead model.

Exit criteria: Contract tests and multiple real adapters pass; statuses distinguish failures.

Phase 4 — Demo hardening

Purpose: Prevent demo behavior from masquerading as production.

Status: Reported implemented; verify.

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Dependencies: Phases 1–3.

Exit criteria: Production smoke tests prove demo actions/data disabled.

Phase 5 — Global discovery improvements

Purpose: Make searches country/category-aware.

Status: Reported partly implemented.

Feature scope:

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Dependencies: Connector framework.

Exit criteria: Tests prove actual provider parameters and honest coverage.

Phase 6 — Safe public discovery

Purpose: Respect public API limits and policies.

Status: Reported partly implemented; policy review ongoing.

Feature scope:

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Dependencies: Phases 3 and 5.

Exit criteria: No production bulk Nominatim; provider can be switched without code rewrite.

Phase 7 — UK and Norway registries

Purpose: Add official registry data sources.

Status: Companies House and Brreg reported implemented; verify.

Feature scope:

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Dependencies: Connector framework and provenance.

Exit criteria: Live/sandbox or fixture tests for partial/malformed/429 responses.

Phase 8 — Security remediation

Purpose: Close high-impact risks before production.

Status: Partially deferred; V4.6.2 and separate SSRF work deferred.

Feature scope:

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Dependencies: All previous phases.

Exit criteria: Security review passes; deferred patch re-evaluated before production.

Phase 9 — International registries and credentials

Purpose: Expand official country coverage and reusable tenant credentials.

Status: Active; ABN credential integration pending review.

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Dependencies: Phases 1–8.

Exit criteria: ABN plus one additional country tested end-to-end; other candidates have go/no-go decisions.

Phase 10 — Global web and directory intelligence

Purpose: Broaden beyond registries with permitted search/directories.

Status: Planned.

Feature scope:

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Dependencies: Phases 2,3,5,8,9.

Exit criteria: Each source has access/retention/attribution documentation; no prohibited scraping.

Phase 11 — Social integrations

Purpose: Add authorized social integrations.

Status: Research-dependent.

Feature scope:

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Dependencies: Credential store, provider approvals, privacy review.

Exit criteria: Each integration has approved use case, scope, quota and policy sign-off.

Phase 12 — Advanced enrichment

Purpose: Improve completeness and explainability.

Status: Planned.

Feature scope:

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Dependencies: Phases 2,9,10,11.

Exit criteria: No fabricated fields; source terms and retention enforced.

Phase 13 — CRM and collaboration

Purpose: Turn leads into a repeatable sales process.

Status: Planned.

Feature scope:

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Dependencies: Lead model and permissions.

Exit criteria: Small business can manage prospect pipeline without another CRM.

Phase 14 — Automated sales agent

Purpose: Automate administration while retaining human control.

Status: Planned.

Feature scope:

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Dependencies: CRM, tasks, audit and compliant channels.

Exit criteria: Observable and reversible; no unapproved outbound send.

Phase 15 — Campaign and outreach

Purpose: Run compliant, measurable campaigns.

Status: Planned; legal/provider review required.

Feature scope:

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Dependencies: Phase 14, provider, market legal review.

Exit criteria: No sends before sender, permission, opt-out and terms checks.

Phase 16 — Subscription and billing

Purpose: Build sustainable plans after costs are understood.

Status: Planned.

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Dependencies: Usage metering and business operations.

Exit criteria: Visible quotas, idempotent payments and tested downgrade behavior.

Phase 17 — Global SaaS infrastructure

Purpose: Scale reliably when measured demand requires it.

Status: Planned.

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Dependencies: Security gate and stable workflows.

Exit criteria: SLOs, recovery objectives, capacity tests and runbooks validated.

Phase 18 — Public launch and growth

Purpose: Launch with trust, support and measurable adoption.

Status: Planned.

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Dependencies: Production readiness and legal review.

Exit criteria: Launch checklist signed; critical bugs fixed; support and rollback ready.

12. Testing, quality gates and release criteria

Test level	Protects	Examples
Unit	Pure logic and edge cases.	ABN checksum, normalization, encryption, scoring, status mapping.
Contract	Connector consistency.	Result shape, missing fields, error mapping, retries, attribution.
Integration	DB/auth/migrations/wiring.	Tenant upsert, migration idempotency, credential retrieval.
Security	Confidentiality and access.	Cross-tenant access, IDOR, SSRF, secret leakage, role bypass.
End-to-end	Real user journeys.	Register → connect → search → dedupe → assign → follow-up → report.
Load/resilience	Stability.	429, timeout, large imports, concurrent jobs, DB pool exhaustion.
Compliance regression	Terms and user controls.	Nominatim limit, Google attribution/storage, opt-out/suppression.

Release gates

- Migration tested on empty DB and representative prior schema.
- Lint, syntax, unit and relevant integration tests pass; blocked tests disclosed.
- Tenant isolation tested using production-like DB role and connection pooling.
- Connectors cover success, empty, partial, malformed, unauthorized, rate-limited, timeout and outage.
- No secrets or unnecessary personal data in logs/errors.
- UI shows real status; no mock success or fabricated counts.
- Backup restore and rollback plan exist.
- Docs list provider requirements, quotas, attribution and limitations.

13. Free-first infrastructure and cost strategy

Area	Free-first direction	Caution
Runtime	Existing Node.js stack and open-source packages.	Hosting, bandwidth and compute may cost money.
Database	PostgreSQL; local/dev free; evaluate hosted free tier carefully.	Free tiers cap storage/connections or sleep.
Queue/cache	Start with simple DB jobs or existing Redis only if needed.	No managed queue without cost/need review.
Registries	Official open data and free government APIs where permitted.	Registration, quotas, license and commercial terms vary.
Web search	Optional adapters; compare free quotas and terms.	Free search quota may not permit persistent redistribution.
Maps/geocoding	Compliant OSM use and provider switching.	Public Nominatim is not a bulk commercial lead backend.
Email	Manual activity log and test mode first.	Production sending needs domain/authentication and may cost.
AI	Optional; local/manual or user-provided key if suitable.	Hosted inference rarely free at scale.
Payments	Delay until pricing validated.	Processing, tax and currency costs vary.
Monitoring	Open-source/self-hosted where practical.	Self-hosting still costs compute and maintenance.

Cost rule: every provider/dependency decision must document free-tier limits, paid trigger, data rights, alternatives, operational burden and removal path.

14. Risks, assumptions and decisions

Risk / decision	Why it matters	Response
API access not guaranteed	Social APIs and registries may require review/partner status.	Provider access matrix before implementation.
Google Places restrictions	Caching/storage rules can conflict with generic persistent lead DB.	Review per-field use; isolate provider-derived data and retention.
Nominatim limits	Public service is low capacity and prohibits systematic POI harvesting.	Use only compliant end-user queries; consider self-hosted extracts for bulk.
Credential architecture incomplete	Research found plaintext config exposure and broad write risks.	Review Phase A, implement Phase B, then migrate connectors.
V4.6.2 deferred	safeFetch/SSRF risks remain a production concern.	Keep visible as production gate; re-evaluate before launch.
Implementation reports unverified	OpenCode report may not match Windows checkout.	Verify branch, commit, diff, tests and hashes.
Data rights	Public does not mean freely reusable or resellable.	Per-source license/retention record and legal review.
Lead accuracy	Registry records may lack contacts or be stale.	Provenance, freshness, missing fields; no false certainty.
Marketing law	Rules differ by country and channel.	Legal review, opt-out and suppression before outreach.
Free-tier changes	Providers can change quota/terms.	Adapters, degradation and provider switching.

Decisions before later phases

- Pick initial customer segment and target markets; global is the direction, not day-one coverage.
- Classify each provider credential as tenant-owned, platform-owned or OAuth-connected.
- Confirm sources permit persistent storage and commercial use.
- Set raw-payload retention and whether raw data is stored at all.
- Decide whether first release supports only business entities or individual contacts too.
- Choose initial outreach channels and applicable compliance controls.
- Measure infrastructure/provider cost before promising free forever.

15. MVP, beta and production launch criteria

MVP — first usable product

- Secure tenant registration/login and tenant isolation.
- Canonical leads with provenance and dedupe.
- CSV/XLSX import/export.
- At least two reliable, compliant discovery sources for a defined market.
- Search filters, lead detail, notes, tags, assignment and simple pipeline.
- Tasks and manual follow-up tracking.
- Honest connector status/error reporting.
- Basic usage limits, logging, privacy notice, source attribution and data deletion/export basics.

Private beta

- Credential management reviewed and complete.
- Additional country registry connectors tested end-to-end.
- Team roles, shared pipeline, saved searches and basic reports.
- Background jobs with tenant quotas and retry visibility.
- Task/reminder automation; no unapproved sends.

- Backup/restore, security testing and support feedback loop.

Production launch

- Security remediation and SSRF controls completed or explicitly risk-accepted.
- Deployment, secrets, monitoring, backups, restore and incident runbooks tested.
- Provider access, licenses, quotas, attribution and commercial use reviewed.
- Compliant outreach only for approved markets/channels.
- Subscription/usage model and support ready if monetization enabled.
- Clear limitations, source disclosures and onboarding.

Recommended strategy: launch a narrow, trustworthy workflow in one or two markets with a small number of permitted sources; prove users can turn discovered leads into customers; then expand based on real demand.

16. Research references

These official references were consulted. Provider terms change; re-check immediately before implementation and launch. They support access and policy constraints, not blanket approval for every use.

Google Maps Platform — Places API policies and attributions
https://developers.google.com/maps/documentation/places/web-service/policies
Attribution and restrictions on prefetching, caching and storing Places content; Place IDs have a specific exception.

OpenStreetMap Foundation — Nominatim Usage Policy
https://operations.osmfoundation.org/policies/nominatim/
Public service limits, identifying User-Agent/Referer, attribution, no systematic POI harvesting and bulk-geocoding restrictions.

OpenStreetMap Foundation — Attribution Guidelines
https://osmfoundation.org/wiki/Attribution_Guidelines
ODbL attribution expectations for public use of OSM-derived data.

Companies House — Get started
https://developer.company-information.service.gov.uk/get-started
Account registration and API authentication.

Companies House — Developer guidelines
https://developer.company-information.service.gov.uk/developer-guidelines
Rate limits and API key security; published default 600 requests per five-minute window.

Brønnøysund Register Centre — Datasets and API
https://www.brreg.no/en/use-of-data-from-the-bronnoysund-register-centre/datasets-and-api/
Official APIs and Norwegian Open Government Data license context.

Brønnøysund — Open Data API docs
https://data.brreg.no/enhetsregisteret/api/dokumentasjon/en/index.html
Search and entity detail API endpoints.

ABN Lookup — Web Services
https://abr.business.gov.au/Tools/WebServices
Web services are free of charge but registration and agreement are required for a GUID.

LinkedIn — Getting access to APIs
https://learn.microsoft.com/en-us/linkedin/shared/authentication/getting-access
OAuth, limited open permissions and approval requirements for many products.

LinkedIn Help — Accessing APIs
https://www.linkedin.com/help/linkedin/answer/a525039
Self-service products and enterprise program access.

OWASP — Multi-Tenant Security Cheat Sheet
https://cheatsheetseries.owasp.org/cheatsheets/Multi_Tenant_Security_Cheat_Sheet.html
Tenant context, isolation, IDOR, cache separation, resource controls, offboarding and audit.

OWASP — API Security Top 10
https://owasp.org/www-project-api-security/
API threat categories including broken authorization, resource consumption, SSRF and unsafe API consumption.

Appendix — Project history baseline

- Repository: mahasubramanyam/Lead_Generation_pipeline; branch reported as feature/global-lead-saas; previously verified reference commit 73a4bf7d6168eb8e273d7c56297a645b853a8ff1.
- Phases 1–3, demo hardening and selected global discovery improvements were reported implemented: multi-tenant foundation, canonical leads/provenance/dedupe, connector framework, CSV/Excel, web-search adapters and connector UI.

- Companies House and Brreg connectors were reported implemented.
- Public Nominatim was reported disabled by default and restricted to explicit development/test opt-in.
- ABN connector was prepared as an isolated patch/review package; credential-store integration remains pending.
- Credential architecture research identified plaintext config exposure, broad writes and unused role checks. Phase A was reported complete with AES-256-GCM store and migration; actual code review is still required.
- V4.6.2 patch and separate SSRF remediation were deferred and remain production-readiness concerns.
- Latvia and other national registries are research candidates, not completed integrations.

Before using this as a live status report, reconcile each status against the actual branch, commit, working tree, tests and deployed environment.